

# React + Node.js Internship Technical Assignment

## Project: Mini School Management System

### Submission Deadline

Submit the complete assignment before **Tomorrow, 12:00 PM (IST)**.

### Objective

Develop a Mini School Management System using React.js and Node.js. The project should demonstrate your understanding of frontend development, backend APIs, authentication, database design, and clean coding practices.

### Technology Stack

**Frontend:** React.js (Vite), React Router, Axios, Material UI / Bootstrap / Tailwind CSS

**Backend:** Node.js, Express.js, MySQL/PostgreSQL, Prisma or Sequelize, JWT Authentication, bcrypt

### Module 1 – Authentication

Create a secure login system for Admin.

- Login using Email and Password
- Store passwords in hashed format
- Generate JWT after successful login
- Protect private routes and APIs
- Logout functionality

### Module 2 – Dashboard

After login, display a dashboard showing:

- Total Students
- Total Teachers
- Total Homework

Create a single API that returns all dashboard information.

### Module 3 – Student Management

Admin should be able to add, edit, delete, search and view students.

**Fields:**

Admission Number (Auto Generated), First Name, Last Name, Class, Section, Parent Name, Parent Mobile, Address.

Implement validation for required fields and unique admission number.

### Module 4 – Teacher Management

Admin should be able to add, edit, delete and view teachers.

**Fields:**

Employee ID (Auto Generated), Name, Email, Mobile Number and Subject.

## Module 5 – Homework Management

Teachers should be able to create, edit, delete and publish homework.

**Fields:**

Homework Title, Description, Subject, Class, Due Date and Attachment.

Allow uploading PDF, Image and Video files. Maximum upload size: 20 MB.

## Module 6 – Parent Portal

Parents should be able to view homework assigned by teachers.

Features:

- View Homework List
- Search Homework
- Filter Homework
- View Homework Details

Parents should only be able to preview PDF, Image and Video files inside the application. No download button should be available.

## Backend Requirements

Use MVC Architecture, REST APIs, JWT Authentication, Middleware, Validation, Centralized Error Handling, Environment Variables and proper database relationships.

## Frontend Requirements

Build a responsive UI with form validation, loading indicators, toast notifications, protected routes, reusable components and a clean folder structure.

## Database

Design relational tables for Users, Students, Teachers and Homework with appropriate relationships.

## Deliverables

Submit the following:

- GitHub Repository
- Demo Video (5–10 minutes) explaining the project and code
- Postman Collection
- SQL Script or Prisma/Sequelize Migration Files
- README.md with setup instructions

## Submission

Send the following via WhatsApp before the deadline:

- GitHub Repository Link
- Demo Video (or Google Drive link)
- Postman Collection (Optional)

## Important Instructions

Write clean and maintainable code. Use meaningful Git commits, validate user inputs, handle errors properly and be prepared to explain your implementation during the technical interview.